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Geochemical Interpretation of In-Situ Carbonated Brine-Sandstone Interaction: Implications for CO₂ Geo-Sequestration

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Abstract

Geo-sequestration of carbon dioxide (CO₂) in underground geological formations is a highly endorsed method to mitigate existing greenhouse gas emissions. During this procedure, CO₂ may dissolve in in-situ fluids (e.g., brine), producing a reactive solute (e.g., carbonated brine). Subsequently, chemical interactions may occur as the solute engages with the host rock, affecting its petrophysical characteristics and CO₂ sequestration capacity. This work reports the findings of an integrated experimental and geochemical modeling investigation aimed at assessing the characteristics of mineral dissolution and potential alterations in the petrophysical properties of Berea Sandstone. A CO₂-saturated brine (i.e., carbonated brine) was injected through Berea sandstone samples at elevated temperature (60 °C) and pressure (20MPa). The main objectives were to evaluate the fluid-rock geochemical reactions (e.g., mineral dissolution) and quantify changes in the porosity and permeability of the rock samples. The experimental study included the effluent brine collection followed by their geochemical analysis. Subsequently, such results were interpreted using geochemical reaction modelling done using the Geochemist's Workbench package. The findings indicated a measurable decrease in post-flood permeability (~5mD ≈5%) due to the dissolution and subsequent precipitation of certain minerals, together with the blocking of pore throats. A little increase in sample porosity (~0.29%) was noted mostly as a result of the breakdown of clay and carbonate minerals. The geochemical reaction modelling revealed that the proper selection of the initial input and set-up parameters is of prime importance, which could improve the quality of the results and lead to a better evaluation of different chemical reactions. The geochemical model indicated that the calcite and albite were dissolved during carbonated brine flooding, and solution pH decreased from 8.5 to ~4.5 with increasing CO₂ concentration in the system, while dawsonite minerals formed as a response to this dissolution. The amount of mineral dissolution could be accurately estimated using geochemical reactive modelling by carefully setting up the controlling parameters of the model. This signifies the complementary nature of experimental work and a well-set numerical simulation model.

Background

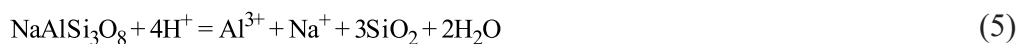
Worldwide, underground gas storage (UGS) is primarily composed of depleted oil/gas reservoirs (over 75%) (Tyne, Barry et al. 2021), followed by aquifer gas storage (approximately 15%) (Marty, Dewonck et al. 2003), salt cavern gas storage (SCGS) (8%) (Li, Liu et al. 2021, Liu, Li et al. 2023, Liu, Li et al. 2024), and abandoned mine gas storage (less than 2%) (Morstyn, Chilcott et al. 2019). Injecting increased accumulates of anthropogenic greenhouse gases, namely, CO₂ into underground geological structures is considered one of the most promising technologies for reducing greenhouse gas emissions (Gunter, Perkins et al. 1993, Bachu, Gunter et al. 1994, Saeedi 2012, Zhang, Wang et al. 2020). Numerous extensive CO₂ storage initiatives have been executed, such as the geological storage in hydrocarbon fields at the Ketzin site in Germany, which is Europe's longest-operating onshore CO₂ storage plant. Approximately 67 kilotons of CO₂ were injected into a saline aquifer, corresponding to about 45 tonnes per day (Schilling, Borm et al. 2009, Würdemann, Möller et al. 2010, Martens, Liebscher et al. 2011, Martens, Kempka et al. 2012, Wiese 2014). The Sleipner CO₂ storage site in the North Sea, initiated by Statoil, exemplifies effective carbon capture with an injection rate of 1 million metric tons of CO₂ annually (Arts, Chadwick et al. 2008, Eiken, Ringrose et al. 2011, Furre, Eiken et al. 2017). Comparable initiatives encompass the Snøhvit site in the Barents Sea and the In Salah field in Algeria (Hansen, Gilding et al. 2013, Ringrose, Mathieson et al. 2013). Depth selection is crucial as it can influence the quantity of stored CO₂, depending on its condition (CO₂ transitions to a supercritical state at temperatures $\geq 31.7^{\circ}\text{C}$ and Pressures $\geq 7.28\text{MP}$). Deep saline aquifers, situated at depths over 800 meters, have been identified as effective reservoirs for CO₂ sequestration (Xu, Apps et al. 2004). Zhang and Arif (2025) claimed that the ideal storage depth for structural trapping decreases with increasing salinity; for example, the maximum CO₂ column height for a selected salinity was determined at a depth of around 1400 m. Accordingly, injected CO₂ could be immobilized through the following trapping mechanisms: structural trapping (Matter, Takahashi et al. 2007, Massarweh and Abushaikha 2024), CO₂ solubility in formation brines (Gunter, Perkins et al. 1993, Yanzhong, Nianmin et al. 2020, Ji, Wang et al. 2024), entrapment by capillarity (Benson and Cole 2008, Li, Yang et al. 2024, Rezk, Adebayo et al. 2024, Al-Bayati, Mahdi et al. 2025), and solidification via chemical reactions with the reservoir rock (Bachu, Gunter et al. 1994, Xu, Apps et al. 2004, Yanzhong, Nianmin et al. 2020, Isah, Mahmoud et al. 2024, Ye, Liu et al. 2025). Both mineral and aqueous trapping convert CO₂ into less mobile forms, hence facilitating permanent sequestration. The latter creates an acidic environment that facilitates the reaction of certain minerals, leading to the formation of new minerals from the liberated cations (Gaus 2010). Nevertheless, the mass transfer between injected CO₂ and the water aquifer is a slow process, and it mainly depends on the fluid/rock interface area (Doughty 2010). Upon the injection of CO₂ into the formation via the injection well, a portion of the CO₂ will dissolve in the formation water, resulting in the creation of carbonic acid (H₂CO₃), as represented by the following reaction equation:



Carbonic acid further dissociates into carbonate ions and hydrogen ions in water:



The hydrogen ions in the formation water will react chemically with the soluble minerals in the reservoir, causing the reservoir minerals to dissolve or new minerals to form, which in turn causes changes in porosity and permeability (Al-Bayati, Saeedi et al. 2019, Tang, Hu et al. 2021). The soluble minerals in the tight sandstone reservoir mainly include Na-feldspar, K-feldspar, dolomite, and calcite (Al-Bayati, Saeedi et al. 2019, Tang, Hu et al. 2021) Their reaction formulas with hydrogen ions are as follows:



Peng, Yang et al. (2022) conducted a feasibility experiment on CO₂ fracturing of tight sandstone, finding that the interaction between tight sandstone and CO₂ increased quartz content in the samples, decreased clay mineral content, led to the formation of new minerals, and caused the dissolution of some minerals on the sample surface. The dissolving of carbonates or feldspars often results in a physical alteration of the core, and this reaction may be observed through the residual solution (Hao, Smith et al. 2013, Smith, Sholokhova et al. 2013). While effective porosity and permeability may increase due to the chemical interaction between rock and fluid resulting in the dissolution of some clay particles, the precipitation of carbonate minerals can markedly diminish these properties. Yu, Liu et al. (2012) assert that particles produced from mineral dissolution may obstruct some pore throats, hence diminishing permeability. The solid precipitation of carbon dioxide is significant since it constitutes a very secure sequestration method. Consequently, comprehending the circumstances under which dissolution and precipitation transpire is crucial for forecasting the development of porosity and permeability, thereby influencing long-term storage mechanisms (Wellman, Grigg et al. 2003, Salem and Shedid 2013). For a more detailed and recent review of the advancement in geochemical and mineralogical research on CO₂/Brine/Rock interactions for CO₂ Sequestration, see recent works (Berrezueta, Kovacs et al. 2023, Fentaw, Emadi et al. 2024, Khan, Siddiqui et al. 2024). In this manuscript, we experimentally investigate the eluent concentrations of core flooding tests and compare and interpret results in the context of a geochemical modeling model. Furthermore, sensitivity analyses were performed using geochemical reactive modeling to investigate the possible amount of reacted CO₂ with the fluid/rock system during injection. The latter is based on observations reported by Saeedi (2012) and Al-Bayati, Myers et al. (2024) in which the residual brine saturation achieved at the end of primary drainage floods using CO₂ (with an IFT of 30mN/m) drainage floods was ~46.4%. Therefore, we hypothesized that setting the reactive porous media with the injected amount of CO₂ to 25, 50, 75, and 100% of the total system pore volume, i.e., CO₂ reactions happen with these percentages within the system, would assist in setting reactive parameters accurately in future studies of reactive transport modelling. Therefore, this could lead to better prediction of CO₂ storage capacity and fluid flow phenomena.

EXPERIMENTAL PROCEDURE AND MATERIAL

A core flooding apparatus, as seen in Fig. 1, was utilized to conduct core flooding testing. Synthetic brine with a density of 1.032 g/mL and a TDS of 38563 was synthesized in the laboratory utilizing distilled water. The concentrations of cations (Na⁺=652.5, Mg⁺⁺=97.4, Ca⁺⁺=12.7, and K⁺=6.3 mmol/L) and anions (Cl⁻=552.1 mmol/L) in the utilised synthetic brine. CO₂-saturated brine was produced via a high-pressure, high-temperature stirred Parr reactor. The Berea sandstone core sample was utilized to conduct core flooding testing. The specimen measures 5.79 cm in length, 3.81 cm in diameter, possesses a porosity of 17.63%, and exhibits a permeability of 98.33 mD. Analysis of XRD results indicated that the core samples comprise the following minerals. 81.2% quartz, 5.5% kaolinite, 4.8% potassium feldspar, 4.5% muscovite, 3% albite, 0.5% dolomite, and 0.3% calcite, all expressed in weight percent. A constant temperature convection oven controlled by a proportional–integral–derivative (PID) controller module was used to regulate the temperature with an accuracy of ±0.5 °C. Two positive–displacement syringe pumps were utilized to control flow rate, volume, and pressure with accuracies of ±0.05 cc/hr, ±0.05 cc, and ±35 KPa, respectively. A special multilayer combination sleeve (aluminium foil, heat shrink Teflon, and one layer of conventional Viton sleeve) was used to ensure Viton sleeve integrity during tests. A relatively low

flow rate (0.5 mL/min) was used to conduct experiments to provide enough time for the injection fluid to react with the rock. A fraction collector was used to collect effluent samples continuously every 10 minutes. Then, laboratory analysis was conducted on these effluent samples to reveal their chemical contents. Refer to Khather, [Saeedi et al. \(2017\)](#) for a detailed description of the equipment, core flooding technique, and protocol. Ion Chromatography (IC) was employed to isolate and quantify ions and polar compounds according to their ionic interactions with a stationary phase. Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) was employed to ascertain the elemental composition of diverse materials. This process entails energizing atoms and ions in a high-temperature plasma, resulting in the emission of light at distinct wavelengths, which are subsequently analysed to identify and quantify the constituents. Finally, Inductively Coupled Plasma Mass Spectrometry (ICP-MS) was employed to detect and measure elements inside a sample, including their isotopes, by ionizing the sample using an inductively coupled plasma and subsequently separating the ions according to their mass-to-charge ratio.

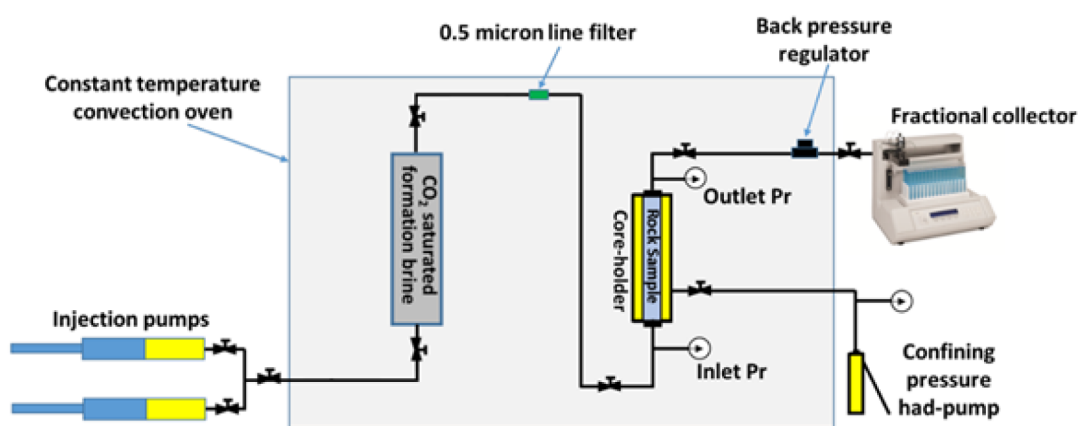


Figure 1—Schematic representation of the core flooding rig configuration.

GEOCHEMICAL MODELLING PARAMETERS AND CONFIGURATION

The potential reactions of rock-fluid and their influences on mineral formation are well described by chemical modeling tools ([Lichtner 1988](#), [Bethke 2008](#)). Geochemical reaction models were used to simulate the results of the core flooding experiment. Static equilibrium models were performed using React (Geochemist's Workbench® Version 11.0) ([Bethke and Yeakel 2013](#)). A reactive transport model divides the region into 100 nodal blocks. Inlet fluid interacts with the medium and discharges at the outlet. The reactive transport modelling may predict reaction front migration over time if flow rates, medium permeability, dispersivity, and reaction rate constants are known ([Lichtner 1988](#), [Bethke 2008](#)). In addition, the reaction model is used to investigate the rates at which minerals dissolve into or precipitate till the system equilibrates by the constraint of kinetic rate laws. In this model, a non-dimensional variable ζ (reaction progress) with the value of 0-1 reflects the total reaction in the model. The modeling aspect of the study was guided by establishing the initial conditions, brine data, mineral composition of the samples, and the porosity and permeability of the samples. The rates of weathering reactions in natural systems are significantly lower than those predicted by models that rely on assumptions about mineral surface area-to-mass ratios ([Black, Carroll et al. 2015](#)). Furthermore, pore and core flooding experiments on CO₂ displacement efficiency suggested the occurrence of CO₂ bypassing due to unfavorable mobility ([Saeedi 2012](#), [Al-Bayati, Myers et al. 2024](#)), combined capillarity and interfacial forces ([Al-Bayati, Mahdi et al. 2025](#)), and heterogeneity influences ([Al-Bayati, Saeedi et al. 2018](#), [Al-Bayati, White et al. 2018](#), [Al-Bayati, Saeedi et al. 2019](#), [Al-Bayati, Saeedi et al. 2019](#), [Al-Bayati, Saeedi et al. 2019](#), [Al-bayati, Saeedi et al. 2022](#)). [Helgeson, Garrels et al. \(1969\)](#) assert that rock progressively interacts with its pore fluid, hence indicating that rock porosity crucially affects the reaction time. Therefore, four models with different amounts of reacted CO₂ are proposed. The

selected amounts of the reacted CO_2 are set to 25, 50, 75, and 100% of the total pore space occupied initially by brine. To initiate modeling, the mineral equilibrium of Ca, Mg, Al, and K was controlled, while the balanced charge of the pore solution and Cl was achieved by adding dissolved Na and pH to the solute. This mixture was brought to equilibrium in correspondence with the time of running the core flooding test (i.e., 1200 minutes). This would facilitate the comparison of the effluent elemental composition derived from core flooding tests with that produced by the geochemical model. Furthermore, it could facilitate the establishment of a thorough comprehension of the resultant data to forecast extended durations of mineral assemblages and transformations.

CORE FLOODING AND GEOCHEMICAL MODELLING RESULTS

The brine eluents gathered in 5 mL plastic vials were tested for quantities of brine elements. IC, ICP-OES, and ICP-MS were used to detect major anions, elements, and trace elements, respectively. Samples were diluted 50-fold in 2% HNO_3 to decrease the salinity to roughly 1000 mg/L. A 2 mL solution was diluted to 20 mL and titrated for HCO_3^- using a standard acid solution (e.g., 0.01 M HCl) to an endpoint of pH 4.3. Ion Chromatography was conducted with a Thermo Dionex ICS 4000 with KOH capillary eluent generation. Samples were diluted 150-fold before analysis to decrease sample salinity. The core flooding data demonstrates that the concentrations of primary ions in solution increase steadily during the experiment. The concentrations of eluent Ca^{++} and Mg^{++} are increased to approximately 150 mg/L and 100 mg/L, respectively, see Fig. 2. Consequently, in a deposit comprising both calcite and dolomite, carbonic acid is likely to react with calcite (Mohamed, He et al. 2011). There was no detectable concentration of eluent Al^{3+} during experimental results analysis, which might be attributable to the scarcity of Al-bearing minerals and the slow hydrolysis rate of feldspar, i.e., experimental time was not enough for mineral dissolution (Xu, Apps et al. 2001). The results indicate a marginal reduction of around 5.0% in the permeability of the examined core sample. A decrease in sample permeability from 98.33 to 93.4 mD was observed. This may be attributed to the dissolution of K-feldspar minerals and the subsequent precipitation of secondary minerals resulting from the availability of cations in the solution (Yang, Wang et al. 2023, Gao, Li et al. 2024). The sandstone samples included in the studies were rich in K-feldspar and albite. In the experiments, calcite and kaolinite precipitated, but albite and K-feldspar underwent partial dissolution. Calcium ions (Ca^{2+}) in formation water are shown to be crucial for the mineral sequestration of carbon dioxide (CO_2). The sustained dissolution of K-feldspar and albite at a gradual rate over an extended duration can extend the period of calcite precipitation and enhance geological CO_2 storage capacity through mineral trapping (Yanzhong, Nianmin et al. 2020). On the other hand, a subtle increase in rock porosity was noticed (i.e., increased by 0.29% of the original porosity).

Geochemical modeling is useful in outlining potential reactions and their impacts on mineralogical formation. They are also crucial in comprehending the outcomes of the core flooding experiment. The geochemical reaction model results show a similar trend, assuming that only about 25-50% of the injected CO_2 reacted with rock minerals (Fig. 3). Geochemical modeling suggests that as the solute pH is reduced from ~8.5 to ~4.5, calcite dissolution occurs; thus, a high eluent concentration of Ca^{++} is formed. The findings align with Wellman, Grigg et al. (2003), indicating that the reaction rate constant of calcite (1×10^{-5} mol/cm²/s) exceeds that of dolomite (1×10^{-7} mol/cm²/s). The findings are confirmed by the reactive model results, which indicate an increased dissolution of calcite accompanied by dolomite precipitation in the system.

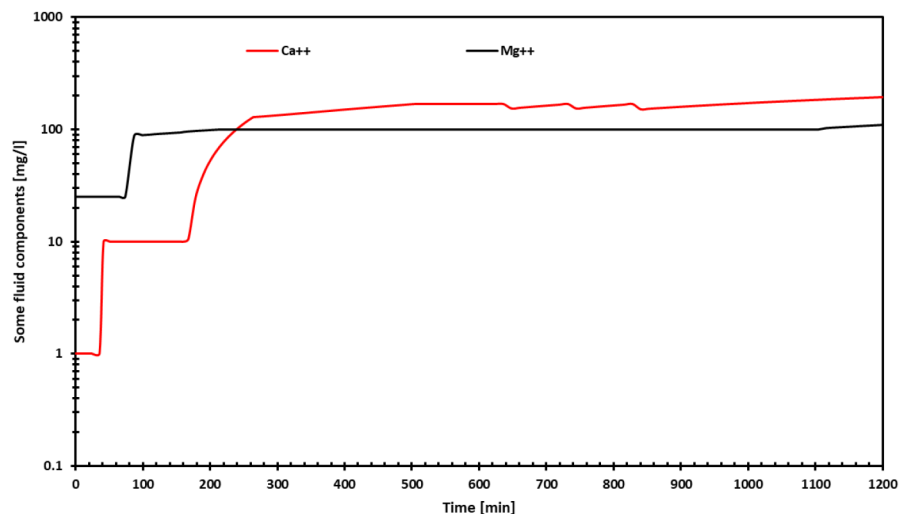


Figure 2—The experimental results of the collected species concentrations of eluent Ca^{++} and Mg^{++} .

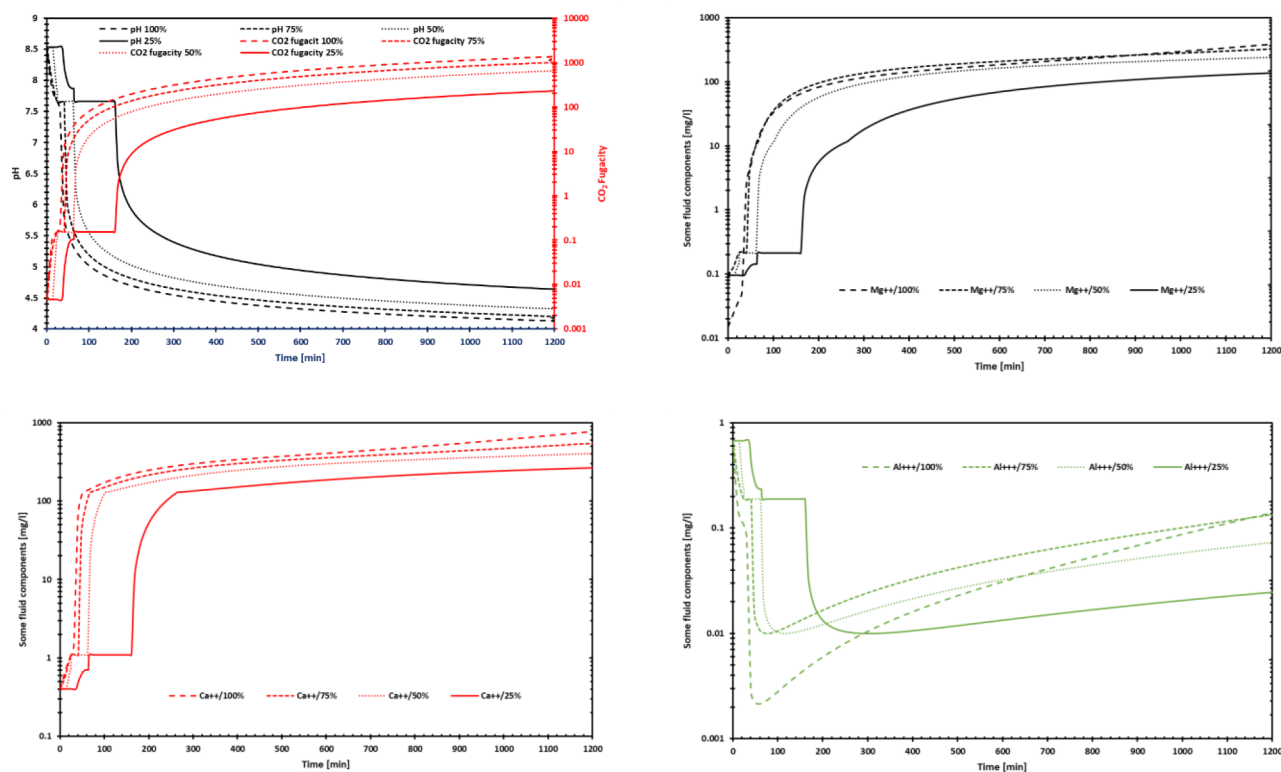


Figure 3—Geochemical modeling results; sensitivity analysis of dissolved components in the brine under different proposed reacted CO_2 amounts, namely, 25, 50, 75, and 100%.

A significant change in mineral and fluid compositions, the reaction in progress, was noticed as CO_2 injection increased (Fig. 4). The formation of carbonic acid from the interaction of CO_2 with brine leads to the possible degradation of minerals, accompanied by a significant reduction in solution pH. CO_2 fugacity initiates at levels near atmospheric values and escalates as additional CO_2 is introduced into the system, indicative of core flooding under elevated pressure, resulting in significant mineral transformation inside the system. Al^{+++} is initially released during the dissolution of clay minerals; thereafter, it decreases as new minerals, such as dawsonite, begin to precipitate. Moreover, the dissolution of albite and calcite facilitated the release of silica (Si) through the side reactions of clay minerals, which led to an increase in quartz and kaolinite in the system. This outcome aligns with the findings of Yang, Wang et al. (2023), indicating a

reduction in the relative proportions of carbonate and feldspar minerals in the tight sandstone, accompanied by an increase in quartz and clay minerals. The concentrations in the core flood experiments are lower than those in the model. The surface area of minerals is frequently one of the least well-defined variables in porous rocks. The kinetic (reactive surface area) may significantly influence the reaction process. Consequently, it introduces significant uncertainty in the outcomes of reactive-transport modeling. In actual systems, weathering reaction rates are significantly lower than those predicted by models that assume specific mineral surface area-to-mass ratios (Black, Carroll et al. 2015). Sensitivity analyses were conducted by decreasing the reactive surface area from 100% to 75%, 50%, and 25%. Results suggested that reactive surface areas of 25% and 50% more accurately represent the reaction process.

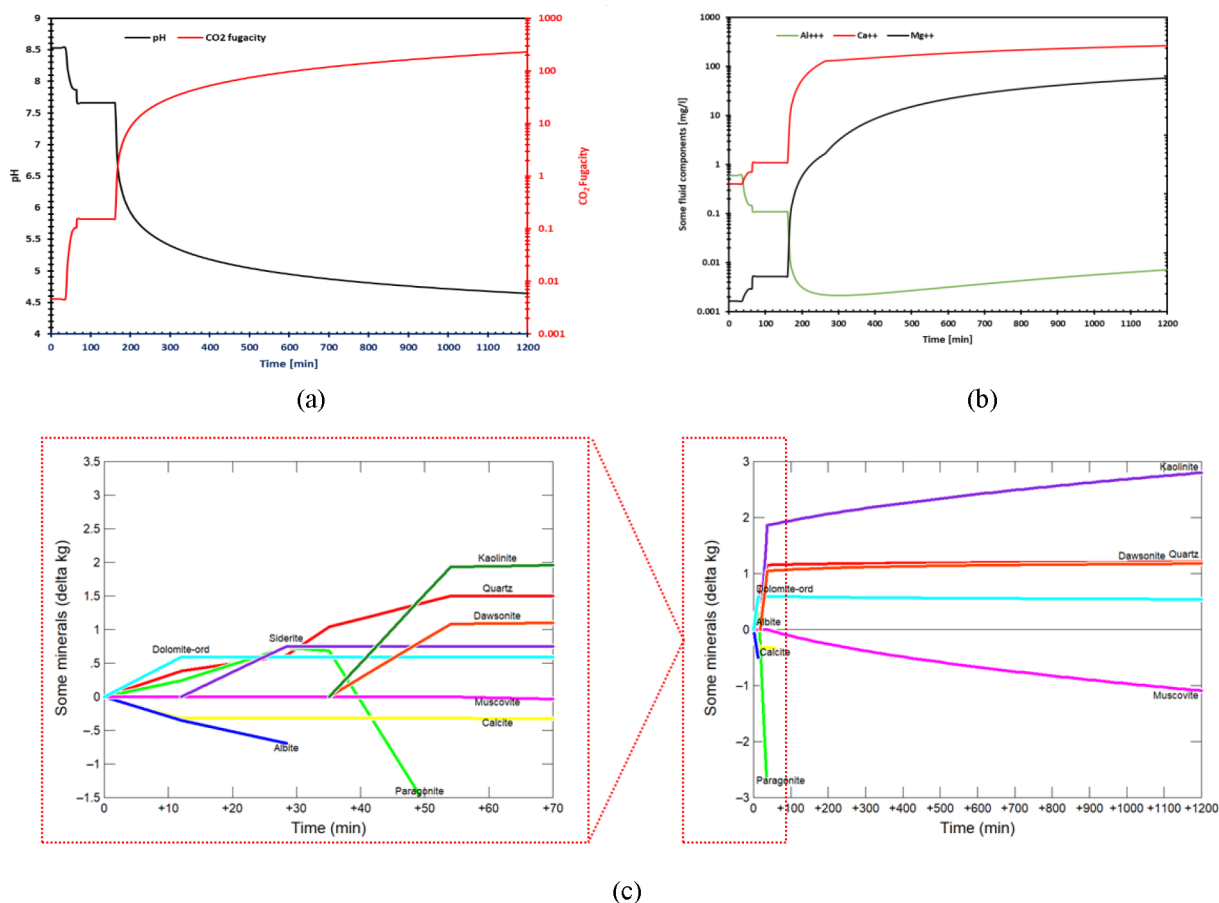


Figure 4—Geochemical modeling results: pH and CO₂ fugacity (a), dissolved components in the brine (b), mineral dissolution/precipitation with time (for 25% reacted CO₂) (c).

Conclusions

Injecting increased accumulates of anthropogenic greenhouse gases, namely, CO₂ into underground geological structures is considered one of the most promising technologies for reducing greenhouse gas emissions. Core flooding studies indicated a quantifiable decrease in post-flood permeability ($\sim 5\text{mD} = \sim 5\%$) due to the precipitation of certain minerals within larger pores (i.e., pore bodies) and the obstruction of pore throats, smaller pores, and dead ends. A slight increase in sample porosity ($\sim 0.29\%$) was mostly attributed to the dissolution of clay and carbonate minerals. The core flooding data demonstrates that the concentrations of primary ions in solution increase steadily during the experiment. The concentrations of eluent Ca⁺⁺ and Mg⁺⁺ are increased to approximately 150 mg/L and 100 gm/L, respectively. There was no detectable concentration of eluent Al³⁺ during experimental results analysis, which might be attributable to the scarcity of Al-bearing minerals and the slow hydrolysis rate of feldspar. Chemical interactions may

arise when a reactive in-situ solution contacts rock. Geochemical modeling studies indicated that around 25-50% of the injected CO₂ interacts with the rock mineral. Geochemical reaction modeling indicates that calcite and albite underwent dissolution, resulting in a fall in solution pH from 8.5 to around 4.5 with the addition of CO₂, while dawsonite and quartz crystals precipitated in response to this dissolution. Al⁺⁺⁺ is initially released during the dissolution of clay minerals; thereafter, it decreases as new minerals, such as dawsonite, begin to precipitate. Moreover, the dissolution of albite and calcite facilitated the release of silica (Si) through the side reactions of clay minerals, which led to an increase in quartz and kaolinite in the system. Therefore, the amount of mineral dissolution could be accurately estimated using geochemical reactive modelling by carefully setting up the controlling parameters of the model. This signifies the complementary nature of experimental work and a well-set numerical simulation model.

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